Siddharth P

AI Systems Engineer | Agentic AI, Local RAG & Edge-to-Cloud Product Engineering

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PROFILE

B.E. AI/ML student focused on building AI-powered software that remains understandable under real product constraints. Public work spans shipped web products, LLM-assisted browser workflows, governed agent prototypes, local-first retrieval, edge sensing, mobile integration, and applied ML research.

TARGET ROLES

AI Systems Engineer | Applied AI Engineer | Agentic AI Engineer | AI Product Engineer | AI Platform Engineer

EXPERIENCE

AI/ML Intern - Equvinoxis Pvt Ltd | Bengaluru | Jun 2025 - Jul 2025

- Supported research and prototyping for AI-assisted facial analysis of pilot stress, including feature extraction, model experimentation, multimodal signal work, and human-lab evaluation.
- Produced experiment documentation to make model comparisons, observations, and limitations reviewable.

FEATURED SYSTEMS

CREDA | Shipped candidate-verification product

Full-stack workflow for structured candidate evaluation, repository analysis, and authenticity signals. Engineering focus: multi-model fallback, Redis/BullMQ background work, Supabase-backed product flows, and the operational decisions around AI-assisted review. Stack: React, Node.js, Supabase, Redis, BullMQ, LLMs.

SmartHelm | Edge-AI rider-safety system

Privacy-first drowsiness detection for delivery riders. A Raspberry Pi helmet unit runs MediaPipe eye analysis with MAX30102 heart-rate/SpO2 sensing and alerts; a Kotlin Android app and Firebase dashboard connect rider, device, and fleet. Stack: Python, Flask, MediaPipe, Raspberry Pi, Kotlin, Firebase.

PayGuard-AgentX | Governed agent prototype

Offline-first retail control-centre prototype for restock, invoice, and fraud-investigation workflows. Uses LangGraph supervision, MCP tool contracts, critics, signed evidence dossiers, and an explicit human decision boundary; not presented as a production-audited system. Stack: Python, LangGraph, MCP, SQLite, Chroma, Kuzu, Ollama.

TrustLayer | Shipped privacy-policy analysis product

Browser and web workflow that converts dense privacy-policy text into an understandable risk signal through an Express-backed, LLM-assisted analysis layer. Stack: Chrome extension, Express, LLM integration.

AARNA | Team-built Algorand testnet

Team-built decentralized MRV workflow for blue-carbon restoration: IPFS project evidence, on-chain validation roles, Algorand Standard Asset credits, and a marketplace model. Stack: Algorand, AlgoPy, Puya, React, TypeScript, IPFS. RIFT 2026; live testnet.

AI SYSTEMS & RESEARCH

NirmiqResearchOS | Local-first academic intelligence MVP
Offline-first academic workspace for source-backed document Q&A, Paper Lab, and Exam Lab. Combines FastAPI, Next.js, SQLite, optional Ollama inference, hybrid retrieval, citations, abstention, CI, and answer-quality evaluation flows. Portfolio/demo MVP, not a hosted multi-user SaaS.

NirmiqCodeSensei | Local developer-learning tool
Local repository analysis tool that provides eight-lens review, architecture mapping, code-grounded DSA, explain-back questions, SQLite storage, and optional MCP integrations. Runs on localhost with no telemetry. Stack: TypeScript, Next.js, SQLite, Drizzle, MCP.

OriginX-T | Dataset-forensics research capstone
Time-aware early-warning research system for contamination risks before LLM fine-tuning damage becomes visible. Scope includes controlled contamination generation, a LoRA fine-tuning grid, collapse-curve measurement, and calibrated risk classification; advanced methods are explicitly gated. Stack: Python, PyTorch, Transformers, PEFT, MLflow.

AscensionOS | Private offline-capable product build
Mobile-first personal operating system with local cache, offline sync queue, optional Supabase sync, PWA behavior, and Android bridge work. Stack: Next.js, TypeScript, Supabase, Vitest, Capacitor.

Vibecode Project Manager | Developer workflow tool
Approval-first, per-project skill selection for Claude Code using deterministic lexical ranking and local configuration writes. Stack: Bash, PowerShell, GitHub Actions.

ADDITIONAL PUBLIC WORK

PostPilot AI - leakage-aware social forecasting study | NirmiqEcho - offline Windows voice assistant and dictation | VyomaGuard - explainable space mission-assurance prototype | Brain Tumour Detection - ResNet, ViT, and Grad-CAM study | Complete public GitHub record

TECHNICAL SKILLS

AI systems: Agent workflows, MCP, LangGraph, tool contracts, human approvals, audit evidence, LLM integration, local RAG, hybrid retrieval, citations, abstention, evaluation workflows

Backend & data: Python, FastAPI, Node.js, Express, REST APIs, Redis, BullMQ, SQLite, PostgreSQL, Supabase, Firebase, Chroma

Product & edge: TypeScript, React, Next.js, Flutter, Kotlin, Android, MediaPipe, Raspberry Pi, HTML/CSS, accessible UI

Delivery: Docker, GitHub Actions, CI/CD, Git, Ollama, Vercel

EDUCATION

B.E. Artificial Intelligence and Machine Learning - B.N.M. Institute of Technology, Bengaluru | 2023 - 2027 | CGPA: 8.61/10